

Jugal Lachhwani

Gujarat, India. (Willing to Relocate)

+91 8866562577 | jugallachhwani15@gmail.com | [Github](#) | [LinkedIn](#) | [Leetcode](#)

Summary

Innovative AI Enthusiast and **2x hackathon award winner** specializing in Generative AI and Agentic AI, with a deep knowledge in Machine Learning, Deep Learning, MLOps and Data Structures. Experienced in building Agentic AI Workflows and end-to-end AI systems, focused on building meaningful projects that can create real world impact. Acknowledged for skills in data engineering and MLOps, adept at streamlining end-to-end machine learning workflows from experimentation to production. Passionate about **building scalable systems and intelligent applications** while continuously learning to adapt to evolving technologies.

Achievements

- **Winner, Odoo x Gujarat Vidyapith Hackathon 2026:** Secured 1st place among 2,200+ national participants by developing a solution of Product Lifecycle Management which ensures the Products and BOM consistency, within 24 hours.
- **2nd Runner-Up, DecodeX National Hackathon:** Secured 3rd rank out of 4,000+ participants in a 24-hour national-level hackathon, building the Stock Market cheating project in less than 24 hours being first among the similar problem statement finalist and 3rd overall.
- **Top 15 among 38,000+ participants in Delhi AI Summit Buildathon:** Ranked within the top 2% among 38,000+ participants across India for innovative AI solution development.
- **Top 8 among the 500+ teams in KenXai Hackathon:** Building the data-Engineering project within 24-hour using tools like Apache Airflow, PostgreSQL and more for building the data models, and robust pipeline.
- **Finalist in Meta x hugging face OpenENV Hackathon:** The Hackathon has over 31,000+ project registrations, and we were among the top 800 teams reached to the Finals
- **Top 50 National Teams, NEF Competition:** Selected as a top 50 team nationwide for a proprietary startup concept in a prestigious competition sponsored by the NEF.
- **Modeling Lead, AIML Robot Initiative:** Successfully secured institutional funding and budget approval from **iCreate** to spearhead the AIML Department's robotics project.

Internship

ISRO

Machine Learning researcher (Internship)

Gujarat, India (Hybrid)

May. 2026 - Nov 2026 (Ongoing)

- Spearheaded end-to-end ML research and development pipelines, leading data preprocessing, model experimentation, and rigorous performance evaluation for real-world predictive applications.
- Engineered a highly accurate time-series forecasting ensemble to predict Kosi River water levels, stacking deep sequence models (LSTM, GRU, Bi-LSTM, RNN) with an XGBoost meta-learner and achieved R^2 score of >0.97 .
- Processed and analyzed geospatial Synthetic Aperture Radar (SAR-I) satellite data to dynamically map flood inundation zones, correlating physical spatial imagery with predictive models.

Samatrix.io

Statistical Analytics (INTERNSHIP)

Gujarat, India (Remote)

May. 2025 - June 2025

- During my internship at Samatrix.io, I gained hands-on experience working as a Data Analyst across diverse domains. I learned to apply various data preprocessing techniques, performed in-depth exploratory data analysis (EDA), and built actionable insights to support business decision-making.
- Worked on datasets from multiple domains such as healthcare, finance, and e-commerce. Applied statistical techniques and data visualization tools (e.g., Python, Pandas, Matplotlib, Seaborn) to identify trends and patterns. Improved my skills in data cleaning, feature engineering, and storytelling with data.

Certifications

NPTEL Data base Management Systems

Langchain Foundation: Introduction to LangGraph – Python used

Anthropic Introduction to Model Context Protocol

Coursera Genrative AI: Prompt Engineering

DeepLearning.AI TensorFlow Developer, Deep Learning, Generative AI with Large Language Models, Convolutional Neural Networks.

IBM Machine Learning with Python

Oracle Oracle Cloud Infrastructure 2025 Certified Generative AI Professional

Skills

Data Science
Programming Languages
Cloud Platforms
Gen and Agentic AI
Frameworks and Libraries

Machine Learning, Deep Learning, MLOps, Computer Vision, NLP
Python, SQL
AWS
LangGraph, LangChain, n8n, RAG, Multi-Agent
Orchestration
PyTorch, TensorFlow, Keras, Scikit-Learn, NumPy, Pandas, Matplotlib, OpenCV,
Seaborn
Docker, Kubernetes, Fast API, Git, GitHub, CI/CD, Apache Kafka
PostgreSQL, Quadrant
DagsHub, MLFlow, DVC
English, Hindi, Gujarati

Development
Databases
MLOps Tools
Languages

Projects

Ecommerce Customer Interaction Agent | Trained using SFT and GRPO

- Architected a stochastic e-commerce simulator via FastAPI with a **22-operation** action space to evaluate LLM agents on long-horizon planning and strict policy compliance.
- Engineered a 4-phase curriculum learning pipeline utilizing Supervised Fine-Tuning (SFT) and Group Relative Policy Optimization (GRPO) to fine-tune a Llama 3B model for complex decision-making.
- Optimized the training architecture using LoRA **4-bit quantization** and **Unsloth**, achieving a 2x training speedup and an **80% reduction in VRAM usage**.
- Designed a multi-signal reward shaping engine and established a robust Supervised Fine-Tuning (SFT) baseline, tweaking Group Relative Policy Optimization (GRPO) feedback giving model to successfully stabilize the mean evaluation reward to **1.07**.

CareerLens

GitHub: [Link](#)

- Architected an **end-to-end multi-agent system using LangGraph** and Python to automate the job search workflow, integrating live data scraping via Apify to match candidates with real-time job postings.
- Developed an intelligent matching engine leveraging LangChain and Llama-3.2-8b local model to parse PDF resumes into structured schemas and calculate a definitive 0-100% ATS compatibility score against live job descriptions.
- Built a real-time analytics pipeline using PostgreSQL and FastAPI to ingest and process market data, powering a dashboard that visualizes top in-demand skills, role density, and geographical market trends.
- Designed a **context-aware GenAI chatbot** utilizing Retrieval-Augmented Generation (RAG) to act as a career mentor, dynamically computing skill gaps and generating custom 30-day upskilling roadmaps grounded in live market analytics.

ScamShield | Scam the Scammers

GitHub: [Link](#)

- **Architected an AI-driven honeypot system** to autonomously detect malicious actors and engage them in deceptive, real-time conversations, achieving a 100% scam detection accuracy rate.
- **Engineered advanced context window management** to handle complex, multi-turn dialogues, ensuring the AI maintains long-term memory to accurately mimic human conversational patterns.
- **Developed a covert backend pipeline** to analyze and extract scammer intelligence and behavioral patterns, securing a 94.3% information extraction accuracy against a custom dataset.
- **Optimized conversational latency** to an average of 5.3 seconds per response, intentionally mirroring natural human typing speeds to prolong engagement and maximize intelligence gathering.

Education

Charotar University of Science and Technology (CHARUSAT)

Gujarat, India

BACHELOR OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING
(Expected)

Sep. 2023 - Apr. 2027

- CGPA: 8.38

Best Higher Secondary School

Gujarat, India

PRIMARY, SECONDARY, AND HIGHER SECONDARY (SCIENCE)

2023

- 10th Grade: 89.21th Percentile
- 12th Grade: 80.41th Percentile

